PAPER ID-310421

Subject Code: KCS054

Roll No: | | | | | | | | | | | | | |

B. TECH.
(SEM V) THEORY EXAMINATION 2020-21
OBJECT-ORIENTED SYSTEM DESIGN

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	Marks	CO
a.	Define Object identify with example.	2	CO1
b.	List the features of Object-oriented paradigms.	2	CO1
c.	Define and Differentiate Link and Association with example.	2	CO2
d.	Define and Differentiate Generalization and Inheritance with example.	2	CO2
e.	How it is different from multiple inheritance and modelled by using nested generalization?	2	CO3
f.	Define and differentiate Procedural and OOP with example.	2	CO3
g.	Write a program to demonstrate the all the keywords used in Exception Handling.	2	CO4
h.	Define friend functions with example.	2	CO4
i.	Define and differentiate virtual and pure virtual functions with example.	2	CO5
j.	Define constructor and destructor with example.	2	CO5

SECTION B

2. Attempt any *three* of the following: **3 x 10 = 30**

Q no.	Question	Marks	CO
a.	What do you understand by Object-Oriented Technology? Discuss the pros and cons of object-oriented technology with suitable example.	10	CO1
b.	What is Data Abstraction? How it is different from encapsulation? Explain with proper example.	10	CO2
c.	Prepare a DFD for computing the volume and surface area of a cone. Inputs are height and the radius of the base of the cone. Outputs are volume and surface area. Discuss some ways of specifying operations.	10	CO3
d.	Discuss the term Link and Association by taking suitable example. Also, define multiplicity.	10	CO4
e.	What is Late Binding & Early Binding? Which technique is used to implement Early and Late binding in Polymorphism? Explain in brief with example.	10	CO5

SECTION C

3. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	What do you understand by a static member function of a class? Discuss their characteristics Give an example where you can justify the use of static member functions.	10	CO1
b.	Why Object-Oriented Programming (OOP) is so important for software industries or in real life? Explain with example. Discuss the pros and cons of object-oriented technology with suitable example.	10	CO

4. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	What is use case driven OOA? How is it different from OOD? Explain OOA process with the help of a diagram in the unified approach.	10	CO
b.	List the properties of a state chart diagram. Draw a state chart for a coin vending machine present at a railway station.	10	CO

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5. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	Differentiate: i. SA/SD and OMT ii. SA/SD and JSD	10	CO
b.	What do we mean by a collaboration diagram? Explain various terms and symbols used in a collaboration diagram. How is polymorphism described using a collaboration diagram? Explain using an example.	10	CO

6. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	What is operator overloading? Differentiate between overloading of binary operator using friend function and without using friend function.	10	CO4
b.	Differentiate overloading and overriding. Write a program to overload "+" operator using friend function to concatenate two strings.	10	CO4

7. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	What does the inheritance mean in C++? What is containership? How does it differ from inheritance? Explain. Write a program to demonstrate how ambiguity is avoided in a single inheritance using scope resolution operators.	10	CO5
b.	What is Pointer in C++? Define const pointers. What is the usage of the pointer in C++? What is the size of a void pointer in C? Demonstrate it.	10	CO5

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PAPER ID-410485

Subject Code: KCS054

Roll No: | | | | | | | | | | | | | |

B.TECH
(SEM V) THEORY EXAMINATION 2021-22
OBJECT ORIENTED SYSTEM DESIGN

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION-A

1. Attempt *all* questions in brief. 2X10=20

a. Differentiate between link and association.
b. What is an abstract class? Is it possible that an abstract class is inherited by another class?
c. Draw a state diagram for electric bulb.
d. When do we use the protected visibility specifier to a class member?
e. When will you make a function inline? Why?
f. Define Destructor. Why do we use a destructor in a program?
g. What is the use of Scope resolution operator?
h. What is a Candidate key?
i. What is the use of friend function in C++?
j. What is information hiding?

SECTION –B

2. Attempt any *three* parts of the following: 10X3=30

a. What is a Constructor? Write down the different characteristics of a constructor. Write a program in C++ for constructor overloading .
b. What is an inline function? Why do we use inline functions in our program? Write a program in C++ for inline function.
c. Explain object oriented programming. What are the main advantages of object oriented programming over procedural programming? Write a program in C++ by creating a class integer and write a function that prints all the prime numbers from the class.
d. Explain all basic concepts of object oriented programming
e. What is an inheritance? Explain the different types of it .Write a program in C++ for multiple inheritance.

SECTION-C

3. Attempt any *one* part of the following: 10X1=10

(a) What is polymorphism? Differentiate between runtime polymorphism and compile time polymorphism.
(b) What is operator overloading? Write a program in C++ for binary operator overloading.

4. Attempt any *one* part of the following: 10X1=10

(a) A farmer wants to cross the river in a boat along with a bag of grass, a goat and a lion. Only one things can be carried in the boat at a time. If the goat is left alone with the grass the grass will be eaten. If the goat is left along with the lion, the goat will be eaten.
(i) Prepare a scenario in which everything is safely transported across the river.
(ii) Prepare the event trace diagram for the above problem.
(b) Define the term multiplicity and quantification with suitable examples.

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5. Attempt any *one* part of the following: 10X1=10

(a) What is UML? List all building blocks of UML. Explain all types of things used in UML .

(b) Explain virtual base class with the help of an example.

6. Attempt any *one* part of the following: 10X1=10

(a) Discuss implicit and explicit type conversion in detail.

(b) Explain generalization, aggregation and association in detail.

7. Attempt any *one* part of the following: 10X1=10

(a) What is friend function in C++? Write a program in C++ to implement friend function.

(b) Explain the following concepts in C++ by taking a suitable example:

(i) This pointer

(ii) Array of objects

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BTECH
(SEM V) THEORY EXAMINATION 2023-24
OBJECT ORIENTED SYSTEM DESIGN

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	Marks	CO
a.	Describe the features of object-oriented languages ?	2	1
b.	Differentiate between structured approach and object oriented approach.	2	1
c.	What is UML ?	2	2
d.	Describe generalization	2	2
e.	What are the three models in OMT ?	2	3
f.	What do you mean by the optimization of design ?	2	3
g.	Write a C++ program to calculate the value of sin (x).	2	4
h.	Explain typecasting in C++..	2	4
i.	Differentiate between public and private member function.	2	5
j.	Explain static data and static function member.	2	5

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a.	What do you understand by object-oriented technology ? Discuss the pros and cons of object-oriented technology with suitable example.	10	1
b.	What do you understand by architectural modeling ? Explain its various concepts and diagrams with suitable example	10	2
c.	What do you mean by documentation? What are the various considerations in documentation designing? Explain.	10	3
d.	Describe briefly the term namespace, identifiers, variables constants, enum.	10	4
e.	Construct a C++ program depicting the concept of multiple inheritance.	10	5

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Discuss the concept of encapsulation with suitable example.	10	1
b.	What do you mean by polymorphism ? Explain it with an example.	10	1

4. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Explain class and object diagrams with examples.	10	2
b.	Prepare a portion of an object diagram for a library book checkout system that shows the date a book is due and the late charges for an overdue book as derived objects.	10	2

5. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Describe the structured analysis and structured design approach with an example.	10	3
b.	How do you map the object-oriented concepts using non-object oriented languages ? Explain with an example.	10	3

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BTECH
(SEM V) THEORY EXAMINATION 2023-24
OBJECT ORIENTED SYSTEM DESIGN

TIME: 3 HRS **M.MARKS: 100**

6.	**Attempt any *one* part of the following:**	**10 x 1 = 10**	
a.	Explain friend function with example.	10	4
b.	Discuss virtual function. How it is different from pure virtual function? Write a program in C++ for it.	10	4

7.	**Attempt any *one* part of the following:**	**10 x 1 = 10**	
a.	Design a class using C++ to create a singly linked list.	10	5
b.	Define constructor. How constructor is different from normal member function. Explain with example.	10	5

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BTECH
(SEM V) THEORY EXAMINATION 2024-25
OBJECT ORIENTED SYSTEM DESIGN WITH C++

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO	Level
a.	Describe the significance of modeling in software engineering.	1	K_2, K_4
b.	Define the conceptual model of UML.	1	K_2, K_4
c.	Create a package diagram for a modular library management system.	2	K_2, K_3
d.	Explain how use case diagrams help in capturing system requirements.	2	K_2, K_3
e.	Describe the role of inheritance adjustment in design optimization.	3	K_2, K_3, K_4
f.	Demonstrate the use of enumerations and constants.	4	K_2, K_3
g.	Explain the use of the this pointer in C++.	5	K_2, K_3

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the concept of encapsulation and information hiding. How do these principles ensure system security and maintainability? Provide a UML example to demonstrate these principles.	1	K_2, K_4
b.	Draw an interaction diagram for a customer service chatbot system. Include time-based interactions and describe how they facilitate system understanding.	2	K_2, K_3
c.	Explain the process of object-oriented analysis and design with a detailed example of a ride-hailing application. Include the steps from requirement gathering to design optimization.	3	K_2, K_3, K_4
d.	Create a C++ program that uses a callback mechanism. Explain how callbacks are implemented in C++ using function pointers or lambda expressions.	4	K_2, K_3
e.	Write a C++ program to implement operator overloading for matrix addition and subtraction. Include detailed comments on the implementation.	5	K_2, K_3

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Compare and contrast object-oriented modeling with structured modeling. Provide a detailed example of a banking system modeled using both approaches.	1	K_2, K_4
b.	Discuss the importance of architecture in object-oriented modeling. Design a UML architecture diagram for a cloud-based file-sharing system.	1	K_2, K_4

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Develop a UML class diagram for a hospital management system. Explain the rationale behind your design choices.	2	K_2, K_3

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(SEM V) THEORY EXAMINATION 2024-25
OBJECT ORIENTED SYSTEM DESIGN WITH C++

TIME: 3 HRS **M.MARKS: 70**

b.	Explain the concept of polymorphism in collaboration diagrams. Design a collaboration diagram for an online learning platform to show polymorphism in accessing different types of course materials.	2	K_2, K_3

5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Write a C++ program to demonstrate the concept of combining three models (class, state, and interaction) in the design of a smart home system.	3	K_2, K_3, K_4
b.	Compare and contrast SA/SD and object-oriented analysis and design. Use a case study of an online bookstore to highlight the differences.	3	K_2, K_3, K_4

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Write a C++ program to demonstrate the use of inline functions. Compare their performance with macros in terms of memory usage and execution speed.	4	K_2, K_3
b.	Create a program in C++ that uses function overloading to calculate the volume of a cube, cylinder, and sphere. Provide detailed commentary on the code.	4	K_2, K_3

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the concept of static data members and functions. Write a C++ program to track the number of objects created for a class.	5	K_2, K_3
b.	Develop a C++ program to implement hybrid inheritance. Explain how ambiguity is resolved using virtual base classes.	5	K_2, K_3

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